

EXP-6 Retrieval of Similar Items using Vector Embeddings (FAISS)

AIM:

To implement a system for retrieving similar movies using high dimensional vector embeddings with FAISS.

DATASET:

MovieLens Dataset (Movie titles)

PROCEDURE:

Step 1: Install and Import Libraries pip install

faiss-cpu sentence-transformers pandas

```
import faiss
import pandas
as pd
from sentence_transformers import SentenceTransformer
```

Step 2: Load Dataset

```
movies = [ "The
Matrix",
           "Inception",
           "Interstellar",
           "The Dark Knight",
           "Titanic",
           "Avatar",
           "The Avengers"
]

df = pd.DataFrame(movies, columns=["title"])
print(df.head())
```


Output:

```
      title
0    The Matrix
1    Inception
2    Interstellar
3    The Dark Knight
4    Titanic
```

Step 3: Load Embedding Model `model =`
`SentenceTransformer('all-MiniLM-L6-v2')`

Step 4: Convert Text to Embeddings

```
embeddings = model.encode(df['title'].tolist())
print(embeddings.shape)
```

Output:
(7, 384)

Step 5: Create FAISS Index

```
dimension = embeddings.shape[1] index
= faiss.IndexFlatL2(dimension)
index.add(embeddings)
```

Step 6: Perform Similarity Search

```
query = "science fiction movie" query_vec =
model.encode([query]) distances, indices =
index.search(query_vec, k=3) results =
df.iloc[indices[0]] print(results)
```


Output:

title
0 The Matrix
2 Interstellar
1 Inception

Step 7: Observations

- FAISS retrieves semantically similar movies
- Embeddings capture meaning, not just keywords
- Results are relevant to query context

Output:

Top 3 similar movies retrieved successfully.

RESULT:

FAISS retrieves similar movies efficiently using high-dimensional embeddings. It enables accurate semantic search for recommendation systems.

